

Chapter 4: Stoichiometry of Chemical Reactions

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PHY201
Lecture 4



Part I

Writing and Balancing Chemical Equations

Outline of Part I: Writing and Balancing Chemical Equations

Chemical Equations

Ionic and Net Ionic Equations

Outline of Part I: Writing and Balancing Chemical Equations

Chemical Equations

Ionic and Net Ionic Equations

What Is a Chemical Equation?

Definition: Chemical Equation

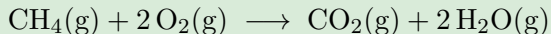
A **chemical equation** uses chemical formulas and symbols to represent a chemical reaction, showing reactants (left side) and products (right side).

Key symbols:

- ▶ $\rightarrow$ reaction arrow (“yields”)
- ▶ $+$ separates multiple reactants or products
- ▶ coefficients — whole-number multipliers placed before each formula

State Symbols

- ▶ (s) — solid
- ▶ (l) — liquid
- ▶ (g) — gas
- ▶ (aq) — aqueous (dissolved in water)



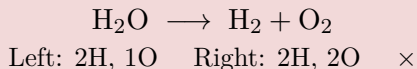
Law of Conservation of Mass

Law of Conservation of Mass

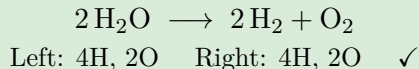
Matter is neither created nor destroyed in a chemical reaction. The total mass of reactants equals the total mass of products.

A chemical equation must be **balanced**: the number of each type of atom must be identical on both sides.

Unbalanced



Balanced

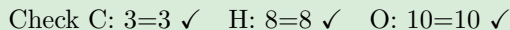
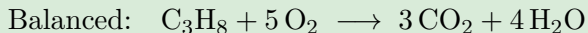
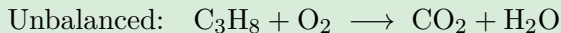


Balancing Equations by Inspection

Strategy

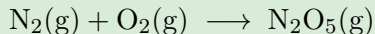
- ▶ Adjust **coefficients** only — never change subscripts (that changes the substance).
- ▶ Balance atoms that appear in the fewest formulas first.
- ▶ Balance H and O last — they appear in many compounds.
- ▶ Verify: atom count matches on every element on both sides.

Example: Combustion of Propane



Example 4.1: Balancing a Chemical Equation

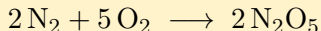
Balance the following equation — (chem2e_ch4_ex4.1)



Step 1: Count as written:

Left: 2N, 2O Right: 2N, 5O

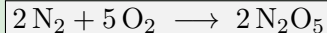
Step 2: Balance O — try $\frac{5}{2}$ on O_2 , then multiply all by 2:



Check:

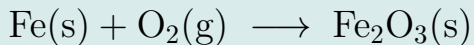
Left: 4N, 10O

Right: 4N, 10O ✓



Check Your Understanding

Balance the following equation:



Come to lecture to see the answer.

Outline of Part I: Writing and Balancing Chemical Equations

Chemical Equations

Ionic and Net Ionic Equations

Three Ways to Write a Reaction

For reactions in aqueous solution, we write equations at three levels of detail.

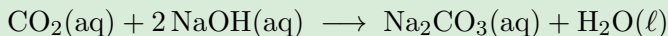
1. **Molecular equation** — complete formulas for all species; no ions shown explicitly
2. **Complete ionic equation** — strong electrolytes written as their dissociated ions
3. **Net ionic equation** — only species that actually change; **spectator ions** are cancelled

Spectator Ions

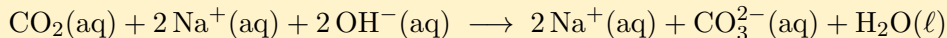
Ions that appear **unchanged** on both sides of the ionic equation; they are present in solution but do not participate in the reaction.

Example 4.2: Writing a Net Ionic Equation

Write the net ionic equation for — (chem2e_ch4_ex4.2)



Complete ionic (split strong electrolytes NaOH and Na₂CO₃):



Net ionic (cancel Na⁺ spectator ions)



Part II

Classifying Chemical Reactions

Outline of Part II: Classifying Chemical Reactions

Precipitation Reactions

Acid-Base Reactions

Oxidation-Reduction Reactions

Other Reaction Types

Outline of Part II: Classifying Chemical Reactions

Precipitation Reactions

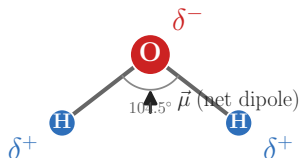
Acid-Base Reactions

Oxidation-Reduction Reactions

Other Reaction Types

Why Does Water Dissolve Ionic Compounds?

Water: A Polar Molecule



Water Is a Polar Molecule

O–H bonds are polar; bent geometry gives a net **dipole moment**: δ^- near O, δ^+ near H.

LIKE DISSOLVES LIKE

Polar/ionic solutes $\leftrightarrow$ polar solvents.

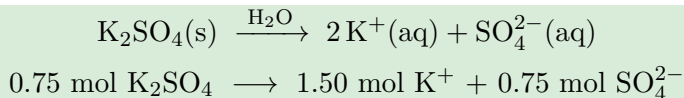
Nonpolar solutes $\leftrightarrow$ nonpolar solvents.

NaCl: water stabilizes Na^+ and Cl^- via **ion–dipole forces (hydration)**.

Ionic Dissociation in Solution

Definition: Dissociation

When an ionic compound dissolves, it separates into its constituent ions. **Strong electrolytes** dissociate completely; **weak electrolytes** only partially.



Ion concentrations are **not equal** to compound molarity! Every mole of K_2SO_4 yields **2 mol K^+** and **1 mol SO_4^{2-}** .

Precipitation Reactions and Solubility Rules

Definition: Precipitation Reaction

Two aqueous solutions are mixed and an **insoluble solid** (the **precipitate**) forms.

Solubility Rules (select):

- ▶ Most NO_3^- , ClO_4^- , $\text{C}_2\text{H}_3\text{O}_2^-$ — **soluble**
- ▶ Most Na^+ , K^+ , NH_4^+ — **soluble**
- ▶ Most Cl^- , Br^- , I^- — soluble (except Ag^+ , Pb^{2+} , Hg_2^{2+})
- ▶ Most SO_4^{2-} — soluble (except Ba^{2+} , Pb^{2+} , Ca^{2+})
- ▶ Most OH^- , S^{2-} , CO_3^{2-} , PO_4^{3-} — **insoluble**

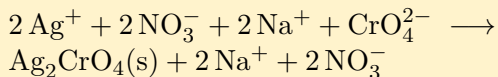
Ion / Salt Type	Solubility	Exceptions
NO_3^- , ClO_4^- , $\text{C}_2\text{H}_3\text{O}_2^-$	Soluble	None
Na^+ , K^+ , NH_4^+	Soluble	None
Cl^- , Br^- , I^-	Soluble	Ag^+ , Pb^{2+} , Hg_2^{2+}
SO_4^{2-}	Soluble	Ba^{2+} , Pb^{2+} , Ca^{2+}
OH^- , S^{2-}	Insoluble	Na^+ , K^+ , Ba^{2+}
CO_3^{2-} , PO_4^{3-}	Insoluble	Na^+ , K^+ , NH_4^+

Example 4.3/4.4: Precipitation of Silver Chromate

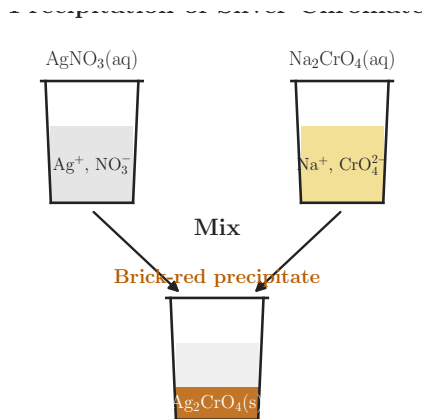
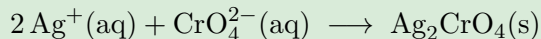
Write the net ionic equation — (chem2e_ch4_ex4.3/4.4)



Complete ionic:



Net ionic



Check Your Understanding

Predict whether a precipitate forms when aqueous BaCl_2 and Na_2SO_4 are mixed. If so, write the net ionic equation.

Come to lecture to see the answer.

Outline of Part II: Classifying Chemical Reactions

Precipitation Reactions

Acid-Base Reactions

Oxidation-Reduction Reactions

Other Reaction Types

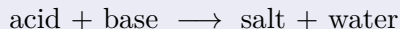
Acid-Base (Neutralization) Reactions

Arrhenius Definitions

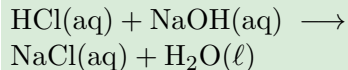
- ▶ **Acid:** produces H^+ (or H_3O^+) in aqueous solution
- ▶ **Base:** produces OH^- in aqueous solution

Neutralization

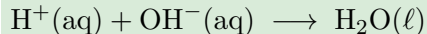
An acid and a base react to form a **salt** and **water**:



Molecular Equation



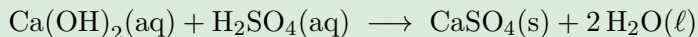
Net Ionic Equation



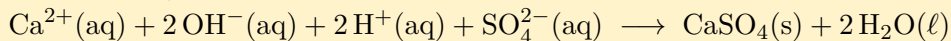
The net ionic equation is **the same** for any strong acid–strong base neutralization.

Example 4.5/4.6: Acid-Base Reactions

Write the net ionic equation — (chem2e_ch4_ex4.5/4.6)



Complete ionic ($\text{Ca}(\text{OH})_2$ and H_2SO_4 are strong; CaSO_4 is *insoluble* — keep as formula unit):



No spectator ions — all ions participate. The complete ionic equation **is** the net ionic equation here.

Outline of Part II: Classifying Chemical Reactions

Precipitation Reactions

Acid-Base Reactions

Oxidation-Reduction Reactions

Other Reaction Types

Oxidation-Reduction (Redox) Reactions

Definition

A **redox reaction** involves the transfer of electrons from one species to another.

Oxidation

Loss of electrons

Oxidation state **increases**

The oxidized species is the **reducing agent**

Reduction

Gain of electrons

Oxidation state **decreases**

The reduced species is the **oxidizing agent**

LEO the lion says GER

Loses **E**lectrons = **O**xidation

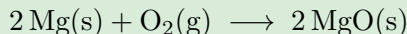
Gains **E**lectrons = **R**eduction

Oxidation Numbers

Rules for Assigning Oxidation Numbers

- ▶ Pure element: oxidation number = 0
- ▶ Monatomic ion: oxidation number = ion charge
- ▶ O in compounds: -2 (except in peroxides: -1)
- ▶ H in compounds: +1 (except in metal hydrides: -1)
- ▶ Sum of oxidation numbers = 0 (neutral compound) or = overall charge (polyatomic ion)

Example: Mg and O₂

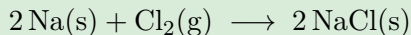


Mg: 0 $\rightarrow$ +2 (oxidized; Mg is the **reducing agent**)

O: 0 $\rightarrow$ -2 (reduced; O₂ is the **oxidizing agent**)

Example 4.7: Identifying Redox

Identify oxidized/reduced species and oxidizing/reducing agents —
(chem2e_ch4_ex4.7)



Na: $0 \rightarrow +1$ **oxidized**

∴ Na is the **reducing agent**

Cl: $0 \rightarrow -1$ **reduced**

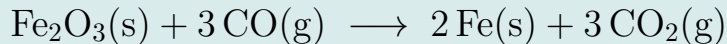
∴ Cl₂ is the **oxidizing agent**

Oxidation and reduction *always* occur together — you cannot have one without the other.

The oxidizing agent is itself **reduced**.
The reducing agent is itself **oxidized**.

Check Your Understanding

Identify the oxidized and reduced species, and state the oxidizing and reducing agents:



Come to lecture to see the answer.

Outline of Part II: Classifying Chemical Reactions

Precipitation Reactions

Acid-Base Reactions

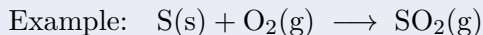
Oxidation-Reduction Reactions

Other Reaction Types

Combination and Decomposition Reactions

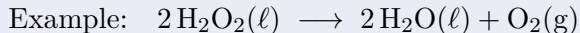
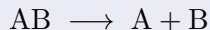
Combination (Synthesis)

Two or more substances combine to form a **single** product:



Decomposition

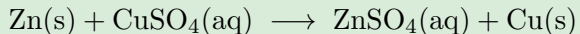
A single compound breaks down into two or more simpler substances:



Single-Displacement Reactions

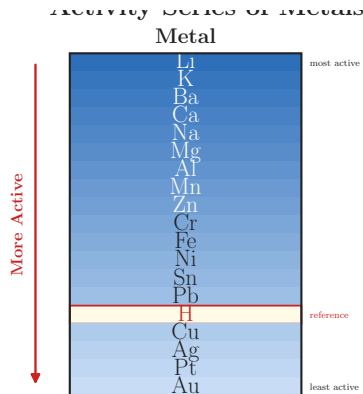
Single-Displacement (Single-Replacement)

One element **replaces** another element in a compound:



Zn: $0 \rightarrow +2$ (oxidized) Cu^{2+} : $+2 \rightarrow 0$ (reduced) — also **redox**.

Displacement requires a **more active** metal to displace a **less active** one (see activity



Activity Series of Metals

Most Active

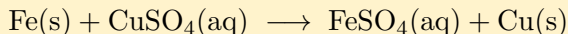
- ▶ Li, K, Ba, Ca, Na
- ▶ Mg, Al, Mn, Zn, Cr
- ▶ Fe, Ni, Sn, Pb
- ▶ **H** (reference)
- ▶ Cu, Ag, Pt, Au

Least Active

A metal **higher** on the list displaces a metal **lower** on the list from its salt solution.

Metals **above H** displace H_2 from acid.

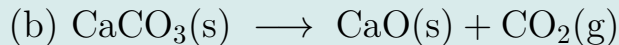
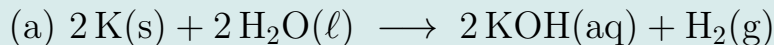
Example: Fe is above Cu, so:



But Cu *cannot* displace Fe^{2+} from $FeSO_4$.

Check Your Understanding

Classify each reaction (may be more than one type):



Come to lecture to see the answer.

Part III

Reaction Stoichiometry and Yields

Outline of Part III: Reaction Stoichiometry and Yields

Stoichiometric Calculations

Limiting Reactant and Reaction Yield

Outline of Part III: Reaction Stoichiometry and Yields

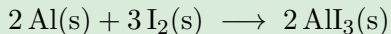
Stoichiometric Calculations

Limiting Reactant and Reaction Yield

The Mole Ratio: Key to Stoichiometry

Mole Ratio

Coefficients in a balanced equation give the **mole ratio** of all reactants and products — the proportional relationship between moles of each species.



Mole ratio $\text{Al} : \text{I}_2 : \text{AlI}_3 = 2 : 3 : 2$

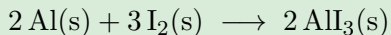
Example conversion factor: $\frac{3 \text{ mol I}_2}{2 \text{ mol Al}}$

Stoichiometric roadmap:

$$\text{mass A} \xrightarrow{\div M_A} \text{mol A} \xrightarrow{\times \text{mole ratio}} \text{mol B} \xrightarrow{\times M_B} \text{mass B}$$

Example 4.8: Moles from Moles

How many mol I_2 react with 0.429 mol Al? — (chem2e_ch4_ex4.8)



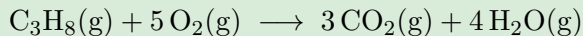
$$n_{\text{I}_2} = 0.429 \cancel{\text{mol Al}} \times \frac{3 \text{ mol I}_2}{2 \cancel{\text{mol Al}}}$$

$$n_{\text{I}_2} = 0.644 \text{ mol I}_2$$

Mole ratio from balanced equation:
$$\frac{3 \text{ mol I}_2}{2 \text{ mol Al}}$$

Example 4.9: Moles to Molecules

How many CO₂ molecules are produced from 0.75 mol C₃H₈? —
(chem2e_ch4_ex4.9)

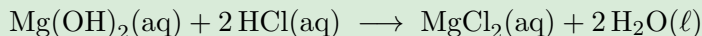


$$N_{\text{CO}_2} = 0.75 \cancel{\text{mol C}_3\text{H}_8} \times \frac{3 \cancel{\text{mol CO}_2}}{1 \cancel{\text{mol C}_3\text{H}_8}} \times \frac{6.022 \times 10^{23} \text{ molecules}}{1 \cancel{\text{mol CO}_2}}$$

$$N_{\text{CO}_2} = 1.4 \times 10^{24} \text{ molecules CO}_2$$

Example 4.10: Mass from Mass

What mass of HCl reacts with 16.0 g Mg(OH)₂? — (chem2e_ch4_ex4.10)



$$m_{\text{HCl}} = 16.0 \text{ g } \cancel{\text{Mg(OH)}_2} \times \frac{1 \text{ mol } \cancel{\text{Mg(OH)}_2}}{58.319 \text{ g}} \\ \times \frac{2 \text{ mol } \cancel{\text{HCl}}}{1 \text{ mol } \cancel{\text{Mg(OH)}_2}} \times \frac{36.461 \text{ g HCl}}{1 \text{ mol } \cancel{\text{HCl}}}$$

$$m_{\text{HCl}} = 20.0 \text{ g HCl}$$

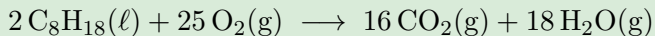
$$M(\text{Mg(OH)}_2) \\ = 58.319 \text{ g/mol}$$

$$M(\text{HCl}) \\ = 36.461 \text{ g/mol}$$

Ratio: 2 mol HCl
per 1 mol Mg(OH)₂

Example 4.11: Large-Scale Stoichiometry

What mass of O_2 is needed to combust 702 g of octane? —
(chem2e_ch4_ex4.11)

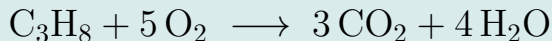


$$m_{\text{O}_2} = 702 \text{ g } \cancel{\text{C}_8\text{H}_{18}} \times \frac{1 \cancel{\text{mol C}_8\text{H}_{18}}}{114.23 \text{ g}} \times \frac{25 \cancel{\text{mol O}_2}}{2 \cancel{\text{mol C}_8\text{H}_{18}}} \times \frac{32.00 \text{ g O}_2}{1 \cancel{\text{mol O}_2}}$$

$$m_{\text{O}_2} = 2.46 \times 10^3 \text{ g O}_2$$

Check Your Understanding

How many grams of CO_2 are produced by the combustion of 44.1 g of C_3H_8 ?



$$[M(\text{C}_3\text{H}_8) = 44.10 \text{ g/mol}; \quad M(\text{CO}_2) = 44.01 \text{ g/mol}]$$

Come to lecture to see the answer.

Outline of Part III: Reaction Stoichiometry and Yields

Stoichiometric Calculations

Limiting Reactant and Reaction Yield

Limiting Reactant

Definitions

- ▶ **Limiting reactant:** completely consumed first; determines the maximum amount of product
- ▶ **Excess reactant:** present in greater than stoichiometric amount; some remains after reaction ends

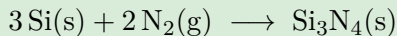
Analogy: Making sandwiches requires 2 bread slices + 1 cheese slice. With 10 bread slices and 4 cheese slices, cheese is the *limiting ingredient* — only 4 sandwiches can be made; 2 bread slices are left over.

Strategy

Calculate the moles of *product* that each reactant could produce independently. The reactant that yields the **fewer moles of product** is the limiting reactant.

Example 4.12: Identifying the Limiting Reactant

2.00 g Si and 1.50 g N₂ are combined. Which is limiting? —
(chem2e_ch4_ex4.12)



From Si:

$$\begin{aligned} n_{\text{Si}_3\text{N}_4} &= \frac{2.00 \text{ g}}{28.086 \text{ g/mol}} \times \frac{1 \text{ mol}}{3 \text{ mol}} \\ &= \boxed{0.02373 \text{ mol}} \end{aligned}$$

From N₂:

$$\begin{aligned} n_{\text{Si}_3\text{N}_4} &= \frac{1.50 \text{ g}}{28.014 \text{ g/mol}} \times \frac{1 \text{ mol}}{2 \text{ mol}} \\ &= \boxed{0.02677 \text{ mol}} \end{aligned}$$

Si gives fewer moles of product ($0.02373 < 0.02677$), so **Si is the limiting reactant.**

Theoretical, Actual, and Percent Yield

Definitions

- ▶ **Theoretical yield:** maximum product calculated from the limiting reactant (assuming 100% conversion)
- ▶ **Actual yield:** amount of product actually obtained in the lab; always $\leq$ theoretical
- ▶ **Percent yield:** the fraction of theoretical yield that was actually recovered

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

Why is actual < theoretical? Side reactions, incomplete conversion, product lost during isolation (filtration, transfer), measurement limitations.

Example 4.13: Percent Yield

1.274 g CuSO_4 reacts with excess Zn; 0.392 g Cu is recovered. Find % yield. — (chem2e_ch4_ex4.13)



$$m_{\text{Cu,theor}} = 1.274 \text{ g } \cancel{\text{CuSO}_4} \times \frac{1 \cancel{\text{ mol CuSO}_4}}{159.609 \text{ g}} \\ \times \frac{1 \cancel{\text{ mol Cu}}}{1 \cancel{\text{ mol CuSO}_4}} \times \frac{63.546 \text{ g Cu}}{1 \cancel{\text{ mol}}}$$

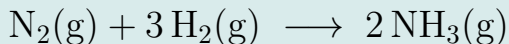
$$m_{\text{Cu,theor}} = 0.5072 \text{ g Cu}$$

$$\% \text{ yield} = \frac{0.392 \text{ g}}{0.5072 \text{ g}} \times 100\%$$

$$\% \text{ yield} = 77.3\%$$

Check Your Understanding

A student mixes 3.00 g N₂ with 0.750 g H₂:



- (a) Which is the limiting reactant?
- (b) What is the theoretical yield of NH₃?

Come to lecture to see the answer.

Part IV

Quantitative Chemical Analysis

Outline of Part IV: Quantitative Chemical Analysis

Titration

Gravimetric and Combustion Analysis

Outline of Part IV: Quantitative Chemical Analysis

Titration

Gravimetric and Combustion Analysis

Quantitative Analysis: An Overview

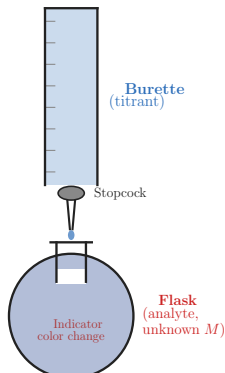
Quantitative chemical analysis uses stoichiometry to determine the amount or concentration of a substance in a sample.

Titration	Gravimetric	Combustion
React sample with a standard solution of known concentration; measure volume needed	Precipitate the analyte; determine mass of dried precipitate	Burn sample; measure CO_2 and H_2O produced

All three methods rely on the same skill: grams $\leftrightarrow$ moles (via molar mass) and mole ratios from a balanced equation.

Titration

Titration Setup



Key Terms

- ▶ **Titrant:** known M , from burette
- ▶ **Analyte:** unknown M , in flask
- ▶ **Equivalence point:** mol titrant = mol needed
- ▶ **Endpoint:** indicator color change
- ▶ **Standard solution:** precisely known M

At equivalence (1:1): $n_{\text{acid}} = n_{\text{base}}$
 General: use balanced-equation mole ratio.

Example 4.14: Acid-Base Titration

50.00 mL HCl requires 35.23 mL of 0.250 M NaOH to reach the endpoint.
Find [HCl]. — (chem2e_ch4_ex4.14)



$$n_{\text{NaOH}} = 0.03523 \text{ L} \times 0.250 \text{ M}$$

$$= 8.808 \times 10^{-3} \text{ mol NaOH}$$

$$n_{\text{HCl}} = n_{\text{NaOH}} = 8.808 \times 10^{-3} \text{ mol}$$

$$[\text{HCl}] = \frac{8.808 \times 10^{-3} \text{ mol}}{0.05000 \text{ L}} = 0.176 \text{ M}$$

1:1 mole ratio

$$n_{\text{HCl}} = n_{\text{NaOH}}$$

$$M = \frac{n}{V}$$

Check Your Understanding

A 25.00 mL sample of H_2SO_4 requires 32.15 mL of 0.150 M NaOH for complete neutralization:



What is the molarity of the H_2SO_4 solution?

Come to lecture to see the answer.

Outline of Part IV: Quantitative Chemical Analysis

Titration

Gravimetric and Combustion Analysis

Gravimetric Analysis

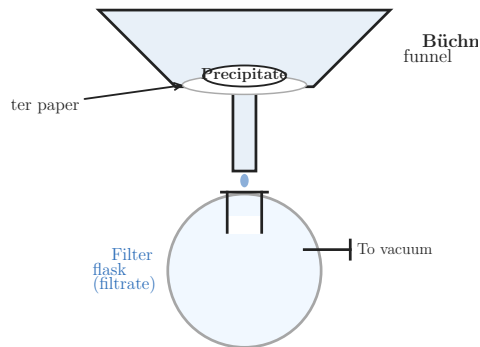
Procedure

- ▶ Dissolve sample; add reagent to **precipitate** the analyte
- ▶ Filter, dry, and **weigh** the precipitate
- ▶ Use stoichiometry to find mass (or %) of analyte

Precipitating reaction must:

- ▶ Be complete (analyte $\approx$ fully precipitated)
- ▶ Give a **known, fixed composition**

vacuum filtration setup



Example 4.15: Gravimetric Analysis

0.4550 g of a $\text{MgSO}_4/\text{MgCO}_3$ mixture gives 0.6168 g BaSO_4 precipitate.
Find % MgSO_4 . — (chem2e_ch4_ex4.15)



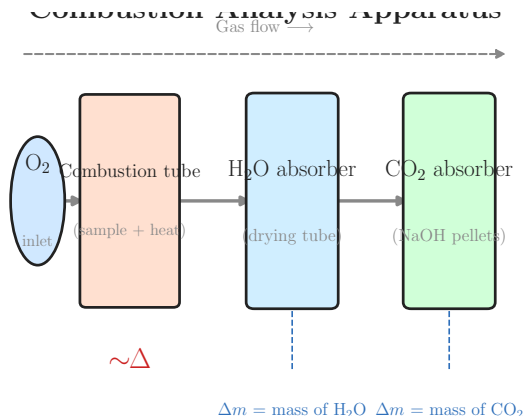
$$\begin{aligned} m_{\text{MgSO}_4} &= 0.6168 \text{ g } \cancel{\text{BaSO}_4} \times \frac{1 \cancel{\text{ mol BaSO}_4}}{233.388 \text{ g}} \times \frac{1 \cancel{\text{ mol MgSO}_4}}{1 \cancel{\text{ mol BaSO}_4}} \times \frac{120.366 \text{ g MgSO}_4}{1 \cancel{\text{ mol}}} \\ &= 0.3181 \text{ g MgSO}_4 \end{aligned}$$

$$\% \text{MgSO}_4 = \frac{0.3181}{0.4550} \times 100\% = 69.91\%$$

Combustion Analysis

Procedure

- ▶ Burn sample in excess O_2
- ▶ Weigh CO_2 (all C) and H_2O (all H)
- ▶ Calculate n_{C} and n_{H} from masses
- ▶ n_{O} by difference
- ▶ Whole-number ratio $\Rightarrow$ empirical formula



Example 4.16: Combustion Analysis

0.00126 g polyethylene $\longrightarrow$ 0.00394 g CO_2 + 0.00161 g H_2O . Find the empirical formula. — (chem2e_ch4_ex4.16)

$$n_{\text{C}} = \frac{0.00394 \text{ g}}{44.009 \text{ g/mol}} \times 1 = 8.953 \times 10^{-5} \text{ mol}$$

$$n_{\text{H}} = \frac{0.00161 \text{ g}}{18.015 \text{ g/mol}} \times 2 = 1.788 \times 10^{-4} \text{ mol}$$

Divide by smallest (n_{C}):

$$\text{C: } \frac{8.953 \times 10^{-5}}{8.953 \times 10^{-5}} = 1$$

$$\text{H: } \frac{1.788 \times 10^{-4}}{8.953 \times 10^{-5}} = 2$$

Empirical formula: $\boxed{\text{CH}_2}$

Check Your Understanding

Combustion of 0.255 g of a compound (C, H, O only) produces 0.561 g CO_2 and 0.306 g H_2O . What is the empirical formula?

Come to lecture to see the answer.

Summary: Chapter 4

- ▶ **Balanced equations** obey conservation of mass; adjust coefficients, never subscripts.
- ▶ **Net ionic equations** omit spectator ions; show only species that change.
- ▶ **Reaction types:** precipitation, acid-base, redox; also combination, decomposition, single-displacement.
- ▶ **Redox:** oxidation = loss of e^- (LEO); reduction = gain of e^- (GER); oxidizing agent is reduced; reducing agent is oxidized.
- ▶ **Stoichiometry roadmap:** mass A $\rightarrow$ mol A $\rightarrow$ mol B $\rightarrow$ mass B via mole ratios.
- ▶ **Limiting reactant:** determines theoretical yield; identify by comparing moles of product from each reactant.
- ▶ **% yield** = (actual/theoretical) $\times$ 100%; always $\leq$ 100%.
- ▶ **Titration:** at equivalence point, apply mole ratio to find unknown concentration.